F. We Were Both Children

time limit per test 3 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: brute force, implementation, math, number theory | https://codeforces.com/problemset/problem/1850/F

Mihai and Slavic were looking at a group of n frogs, numbered from 1 to n, all initially located at point 0. Frog i has a hop length of a_i.

Each second, frog i hops a_i units forward. Before any frogs start hopping, Slavic and Mihai can place **exactly one** trap in a coordinate in order to catch all frogs that will ever pass through the corresponding coordinate.

However, the children can't go far away from their home so they can only place a trap in the first n points (that is, in a point with a coordinate between 1 and n) and the children can't place a trap in point 0 since they are scared of frogs.

Can you help Slavic and Mihai find out what is the maximum number of frogs they can catch using a trap?

Input

The first line of the input contains a single integer t ($1 \le t \le 100$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the number of frogs, which equals the distance Slavic and Mihai can travel to place a trap.

The second line of each test case contains n integers $a_1, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the lengths of the hops of the corresponding frogs.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case output a single integer — the maximum number of frogs Slavic and Mihai can catch using a trap.

Example

Input

```
7
5
1 2 3 4 5
3
2 2 2
6
3 1 3 4 9 10
9
1 3 2 4 2 3 7 8 5
1
10
8
7 11 6 8 12 4 4 8
10
9 11 9 12 1 7 2 5 8 10
```

Output
3 3 3 5 0 4 4

Note

In the first test case, the frogs will hop as follows:

- Frog 1: 0 → 1 → 2 → 3 → **colorred4** → …
- Frog 2: 0 → 2 → **colorred4** → 6 → 8 → …
- Frog 3: 0 → 3 → 6 → 9 → 12 → …
- Frog 4: 0 → **colorred4** → 8 → 12 → 16 → …
- Frog 5: 0 → 5 → 10 → 15 → 20 → …

Therefore, if Slavic and Mihai put a trap at coordinate 4, they can catch three frogs: frogs 1, 2, and 4. It can be proven that they can't catch any more frogs.

In the second test case, Slavic and Mihai can put a trap at coordinate 2 and catch all three frogs instantly.